



ITALIAN STATA CONFERENCE

Triple Differences

*A design for when parallel trends breaks
down — at the price of a new assumption*

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Where we are: parallel trends has broken down

- Everything so far has rested on one assumption: the untreated potential outcome $Y(0)$ would have trended the same way in treated and comparison groups.
- We have spent the session *evaluating* that assumption — pre-trends, event studies, falsifications.
- **Suppose the evaluation fails.** You have good reason to believe the DiD is biased.

What then?

One answer — not the only one — is to find a second, equally biased diff-in-diff and subtract it.

The trade: you relax one assumption and take on another

Difference-in-differences

Requires: $Y(0)$ trends are parallel across groups.

Triple differences

Does *not* require parallel trends. Requires instead: **the bias is the same in both diff-in-diffs.**

- This is **not** a weaker assumption — it is a *different* one, and in some settings a stronger one.
- You are asserting something about the *structure of the confounding*, not its absence.
- It is testable in the same way PT is: event-study plots, now on all three series.

DDD is a research design, not a falsification.

LESSON

Triple Differences

A real design, with one parallel-bias assumption

Triple differences is a research design

- Many people equate triple differences with falsification exercise, but actually it isn't that – it is its own design
- You use triple differences when you have a parallel trends violation – that is, when your diff-in-diff is biased
- Triple differences may sound like a falsification, but it isn't – it's a research design you use when parallel trends is violated
- Miller, Johnson and Wherry (2021) didn't use triple differences with near elderly because they didn't think they had a parallel trends violation – they used a falsification

Biased diff-in-diff #1

Table: Biased diff-in-diff #1: comparing states

States	Period	Outcomes	D_1	D_2
Experimental states	Before	$Y = NJ$	$NJ_t + D$	$D + (NJ_t - PA_t)$
	After	$Y = NJ + NJ_t + D$		
Non-experimental states	Before	$Y = PA$	PA_t	$D + (NJ_t - PA_t)$
	After	$Y = PA + PA_t$		

$$\hat{\delta}_{did}^{true} = D + (NJ_t - PA_t)$$

The ATT is D. Assume, though, that parallel trends does not hold, $(NJ_t \neq PA_t)$

Biased Placebo diff-in-diff

Table: Biased placebo diff-in-diff: comparing states but single men and older women

States	Period	Outcomes	D_1	D_2
Experimental states	Before	$Y = NJ$	NJ_t	$(NJ_t - PA_t)$
	After	$Y = NJ + NJ_t$		
Non-experimental states	Before	$Y = PA$	PA_t	
	After	$Y = PA + PA_t$		

$$\hat{\delta}_{did}^{placebo} = (NJ_t - PA_t)$$

Assume that parallel trends does not hold, $(NJ_t \neq PA_t)$

Two biased diff-in-diffs

- Parallel trends does not hold, ($NJ_t \neq PA_t$), but what if that's the same bias in our placebo DiD?
- Then we can subtract the second from the first:

$$\hat{\delta}_{ddd} = \hat{\delta}_{did}^{true} - \hat{\delta}_{did}^{placebo}$$

- Triple differences is a “real design” with one parallel trends assumption:

$$(NJ_t^{true} - PA_t^{true}) = (NJ_t^{placebo} - PA_t^{placebo})$$

Triple differences by Gruber (1995)

TABLE 3—DDD ESTIMATES OF THE IMPACT OF STATE MANDATES
ON HOURLY WAGES

Location/year	Before law change	After law change	Time difference for location
A. Treatment Individuals: Married Women, 20–40 Years Old:			
Experimental states	1.547 (0.012) [1,400]	1.513 (0.012) [1,496]	– 0.034 (0.017)
Nonexperimental states	1.369 (0.010) [1,480]	1.397 (0.010) [1,640]	0.028 (0.014)
Location difference at a point in time:	0.178 (0.016)	0.116 (0.015)	
Difference-in-difference:	– 0.062 (0.022)		
B. Control Group: Over 40 and Single Males 20–40:			
Experimental states	1.759 (0.007) [5,624]	1.748 (0.007) [5,407]	– 0.011 (0.010)
Nonexperimental states	1.630 (0.007) [4,959]	1.627 (0.007) [4,928]	– 0.003 (0.010)
Location difference at a point in time:	0.129 (0.010)	0.121 (0.010)	
Difference-in-difference:	– 0.008: (0.014)		
DDD:	– 0.054 (0.026)		

Triple differences commentary

- Some people think that it requires that the placebo DiD be zero, but that's incorrect
- In Gruber's 1995 article, it isn't clear why he needed triple differences in the first place – his triple differences yielded -0.054 which is almost the same as what he found with his first diff-in-diff (-0.062)
- The main value of triple differences is that you use it when you believe the parallel trends assumption doesn't hold

Table: Difference-in-Difference-in-Differences (Gruber version)

Groups	States	Period	Outcomes	D_1	D_2	D_3
Married women 20-40	Experimental states	After	$NJ + MW + \textcolor{blue}{NJ}_t + \textcolor{red}{MW}_t + D$	$\textcolor{blue}{NJ}_t + MW_t + D$	$D + \textcolor{blue}{NJ}_t - PA_t$	D
		Before	$NJ + MW$			
	Non-experimental states	After	$PA + MW + PA_t + MW_t$	$PA_t + MW_t$		
		Before	$PA + MW$			
Single men Older women	Experimental states	After	$NJ + SO + NJ_t + SO_t$	$NJ_t + SO_t$		
		Before	$NJ + SO$			
	Non-experimental states	After	$PA + SO + PA_t + SO_t$	$PA_t + SO_t$		
		Before	$PA + SO$			

Triple diff assumption

$$\hat{\delta}_{DDD} = D + \underbrace{[(NJ_t^{MW} - PA_t^{MW}) - (NJ_t^{SO} - PA_t^{SO})]}_{\text{Equally biased DiD \#1 and \#2}}$$

Triple differences requires two diff-in-diff, from different groups, with the same bias. Parallel bias

DDD in Regression

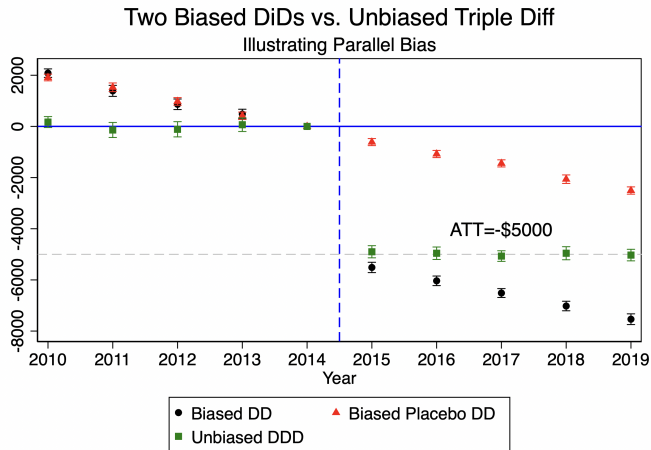
$$Y_{ijt} = \alpha + \beta_2\tau_t + \beta_3\delta_j + \beta_4D_i + \beta_5(\delta \times \tau)_{jt} \\ + \beta_6(\tau \times D)_{ti} + \beta_7(\delta \times D)_{ij} + \beta_8(\delta \times \tau \times D)_{ijt} + \varepsilon_{ijt}$$

- Your dataset will be stacked by group j and state i
- $\widehat{\beta}_8$ estimates the ATT
- Parallel bias, NA and SUTVA necessary and sufficient for identification

Simulation

In /Labs/DDD I have a simulation to illustrate this for us called ddd2.do. The ATT is -\$5,000 but the biased DiD is -\$7487. The non-parallel trends bias is -\$2,487. So I replicate Gruber (with simulated data) where the placebo DiD is close (-\$2,507). I then present a triple differences which gives us -\$4,972. Let's look at the final product.

Triple differences event study





The triple difference estimator

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Summary: Triple difference has become a widely used estimator in empirical work. A close reading of articles in top economics journals reveals that the use of the estimator to a large extent rests on intuition. The identifying assumptions are neither formally derived nor generally agreed on. We give a complete presentation of the triple difference estimator, and show that even though the estimator can be computed as the difference between two difference-in-differences estimators, it does not require two parallel trend assumptions to have a causal interpretation. The reason is that the difference between two biased difference-in-differences estimators will be unbiased as long as the bias is the same in both estimators. This requires only one parallel trend assumption to hold.

Keywords: DID, DDD, DTD, DIDDT, difference-in-difference-in-differences, difference-in-differences, parallel trend assumption, triple difference.

JEL Codes: C10, C18, C21.

1. INTRODUCTION

The triple difference estimator is widely used, either under the name 'triple difference' (TD) or the name 'difference-in-difference-in-differences' (DDD), or with minor variations of these spellings. Triple difference is an extension of double differences and was introduced by Gruber (1994). Even though Gruber's paper is well cited, very few modern users of triple difference credit him for his methodological contribution. One reason may be that the properties of the triple difference estimator are considered obvious. Another reason may be that triple difference was little more than a curiosity in the first ten years after Gruber's paper. On Google Scholar, the annual number of references to triple difference did not pass one hundred until year 2007. Since then, the use of the estimator has grown rapidly and reached 928 unique works referencing it in the year 2017.¹

Looking only at the core economics journals *American Economic Review* (AER), *Journal of Political Economy* (JPE), and *Quarterly Journal of Economics* (QJE), we have found 32 articles using triple difference between 2010 and 2017, see Table A1 in Appendix A. A close reading of these articles reveals that the use of the triple difference estimator to a large extent rests on

¹ More details on the historical development of the use of the triple difference estimator can be found in the working paper version of Olden and Møen (2020, fig. 1). In the working paper, we also analyse naming conventions and suggest that there is a need to unify terminology. We recommend the terms 'triple difference' and 'difference-in-difference-in-differences'.

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Summarizing DDD

- Used to be people thought DDD required two parallel trends assumptions but it does not – it is a real design and requires one parallel trends assumption
- Parallel trends assumption is “parallel bias” – that the bias of the true DiD is the same as the bias of the placebo DiD
- The ladder of evidence still holds – you’ll want to present the event study plot, and my code provides it for you, because you need to evaluate the parallel bias assumption
- Given the lack of triple diff literacy, you may have to write this anticipating reader and maybe editor confusion and so “educate” as you go – overlaying all three plots could be help